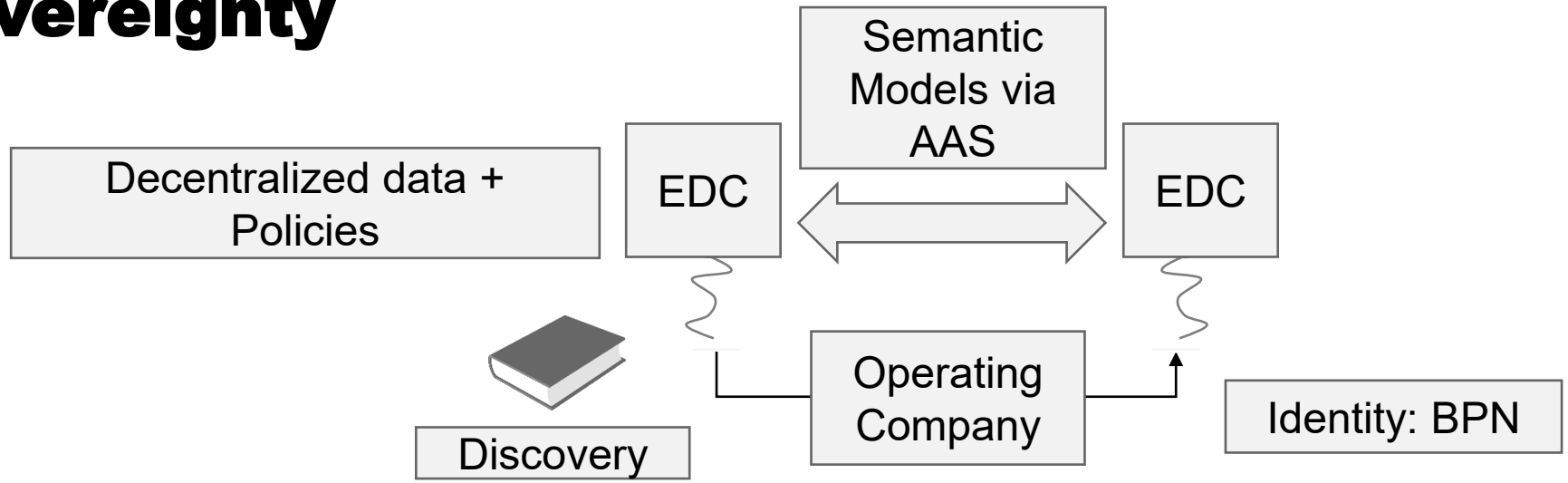
From Catena-X to Cross-Industry Adoption: A Practical Guide to Joining the Eclipse Tractus-X Community

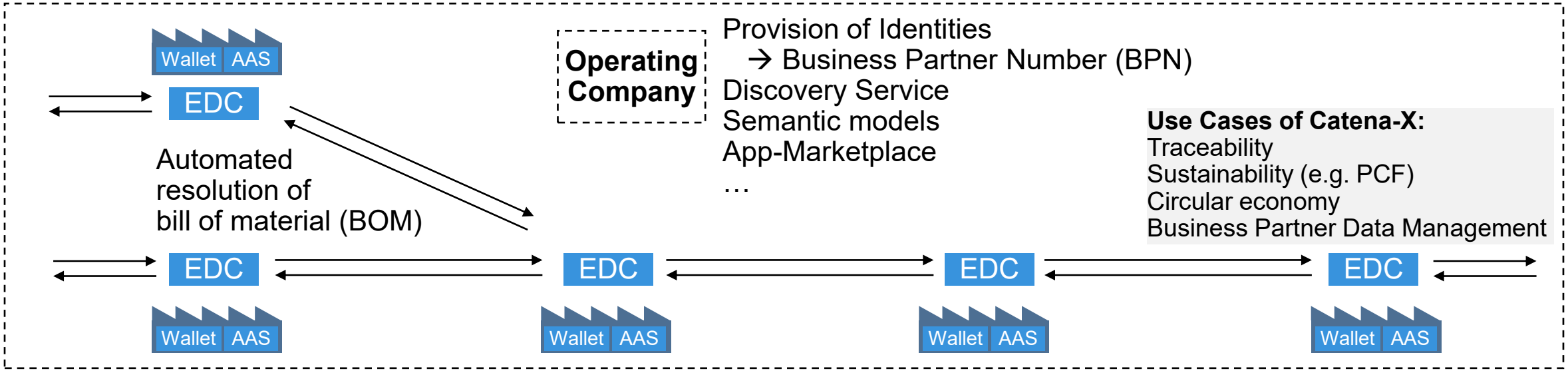
David Görzig

ARENA2036 e.V.

ECLIPSE Tractus-X™
Where we build dataspaces!

Dataspaces – technical implementation of data sovereignty

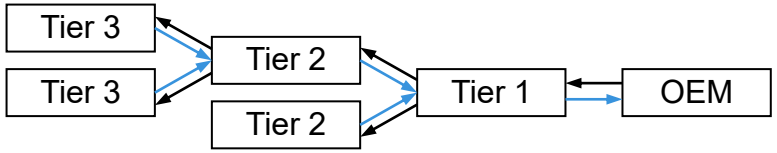




Example use cases in the Catena-X universe

Product Carbon Footprint

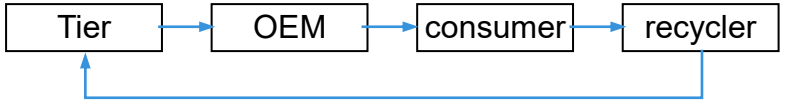
PCF-requests in value chain



Automatic PCF-provisioning

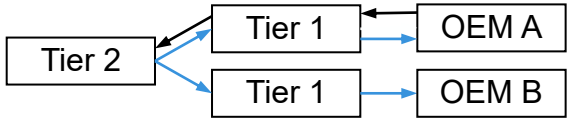
Circular economy (Product Pass)

Provisioning of product related data



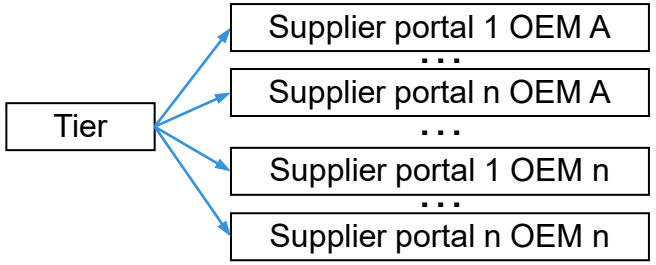
Traceability (e.g. Quality)

Quality requests in value chain



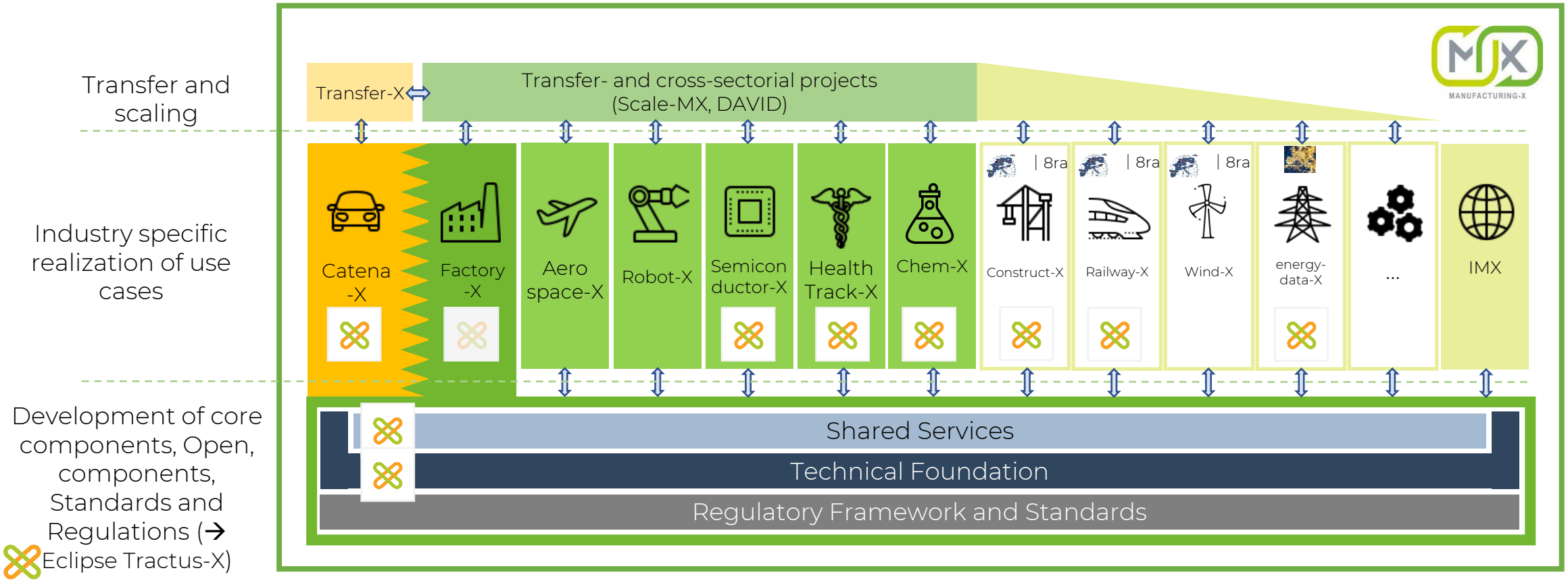
Automatic data provisioning

Business Partner Data Management (company data, certificates, …)



Data spaces were developed in order to realize cross-company use cases

Tractus-X the common approach on Manufacturing-X

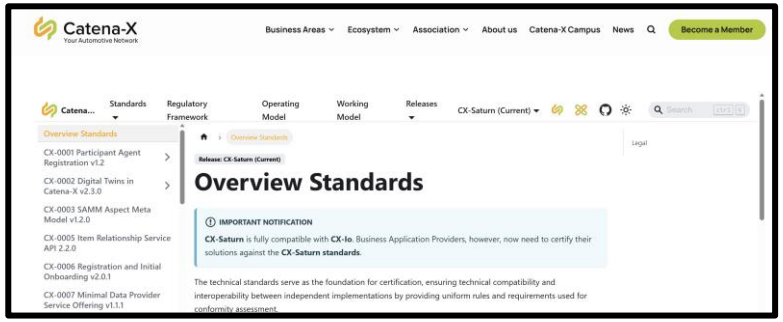


Phase 1 – Learn about Tractus-X

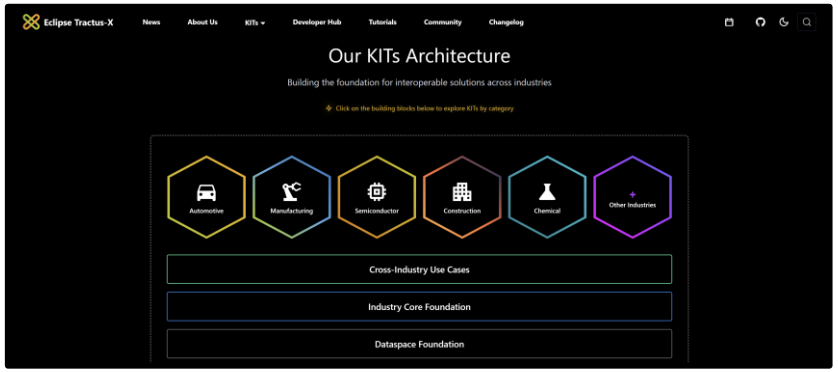
- Catena-X Standards: The non-negotiable normative baseline. They strictly dictate system architecture and data semantics.

- Tractus-X KITs: The exact reference implementations for specific data space use cases. They serve as the absolute Single Source of Truth.

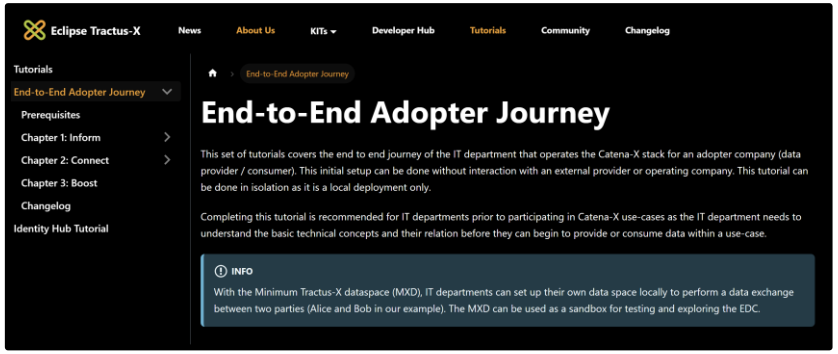
- End-to-End Adopter Journey: Mandatory hands-on deployment (via Kubernetes/Docker). Theoretical knowledge without local system execution is categorically insufficient for contributing.



https://catena-x.net/association/standardization/



https://eclipse-tractusx.github.io/Kits/



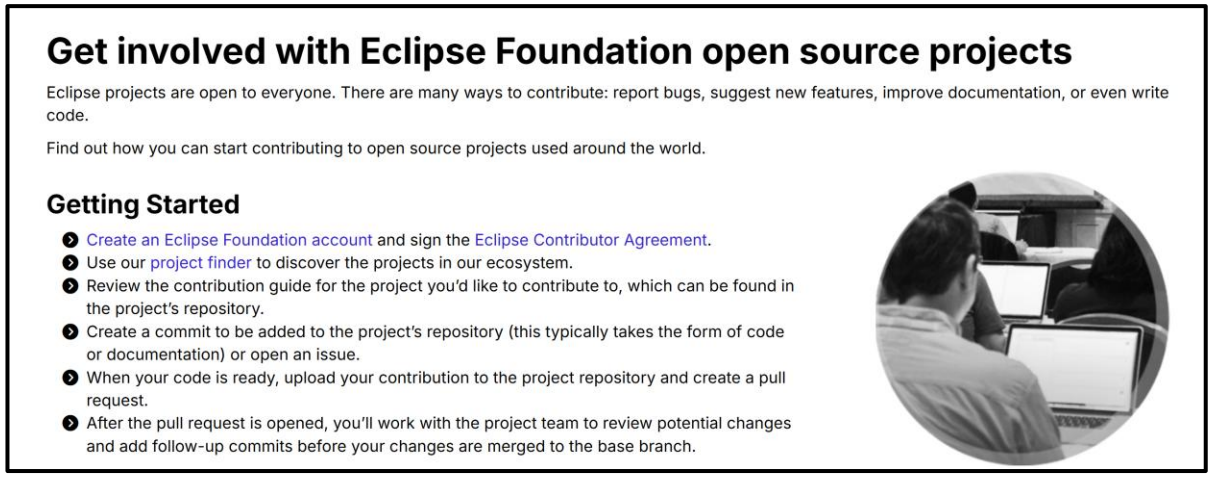
https://github.com/eclipse-tractusx/tractus-x-umbrella/tree/main/docs/user/

Phase 2 – Registration

- Eclipse Foundation Account: Registration utilizing a corporate email address is mandatory.

- GitHub Integration: Your GitHub account must be strictly linked to the exact Eclipse email address.

- Eclipse Contributor Agreement (ECA): Requires a digital signature to guarantee intellectual property compliance.



Get involved with Eclipse Foundation open source projects

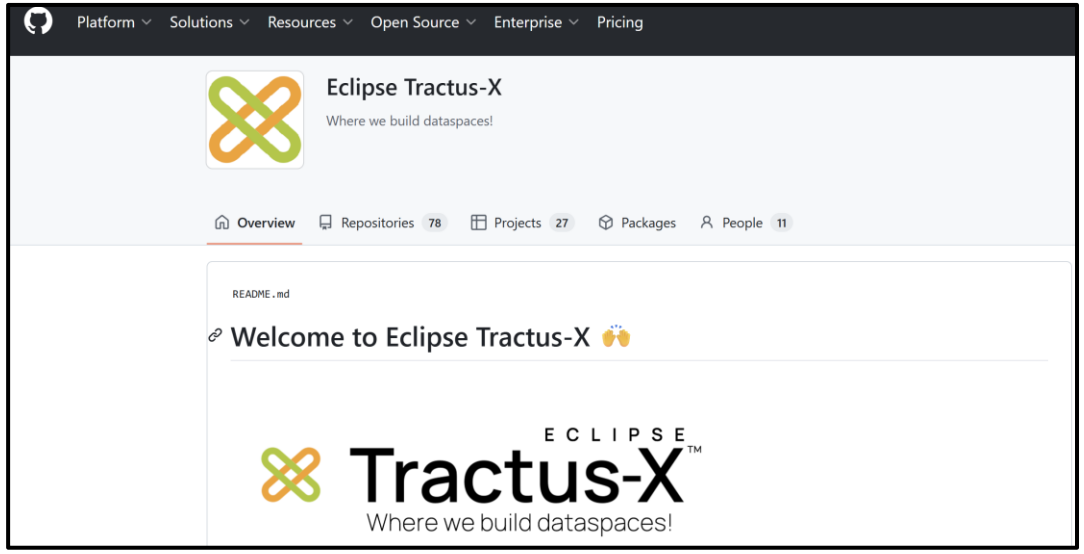
Eclipse projects are open to everyone. There are many ways to contribute: report bugs, suggest new features, improve documentation, or even write code.

Find out how you can start contributing to open source projects used around the world.

Getting Started

1. Create an Eclipse Foundation account and sign the Eclipse Contributor Agreement.
2. Use our project finder to discover the projects in our ecosystem.
3. Review the contribution guide for the project you'd like to contribute to, which can be found in the project's repository.
4. Create a commit to be added to the project's repository (this typically takes the form of code or documentation) or open an issue.
5. When your code is ready, upload your contribution to the project repository and create a pull request.
6. After the pull request is opened, you'll work with the project team to review potential changes and add follow-up commits before your changes are merged to the base branch.

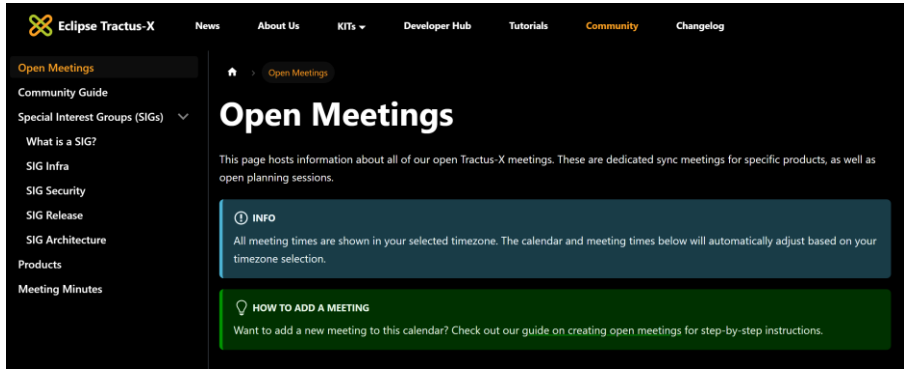
https://www.eclipse.org/contribute/



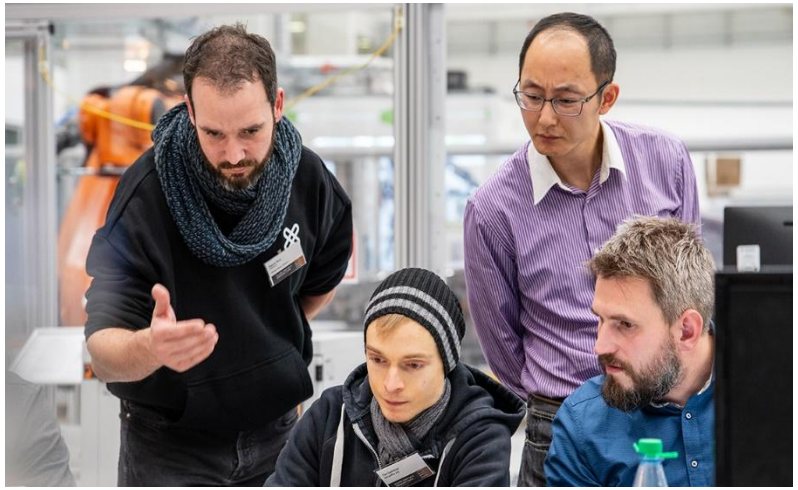
https://github.com/eclipse-tractusx

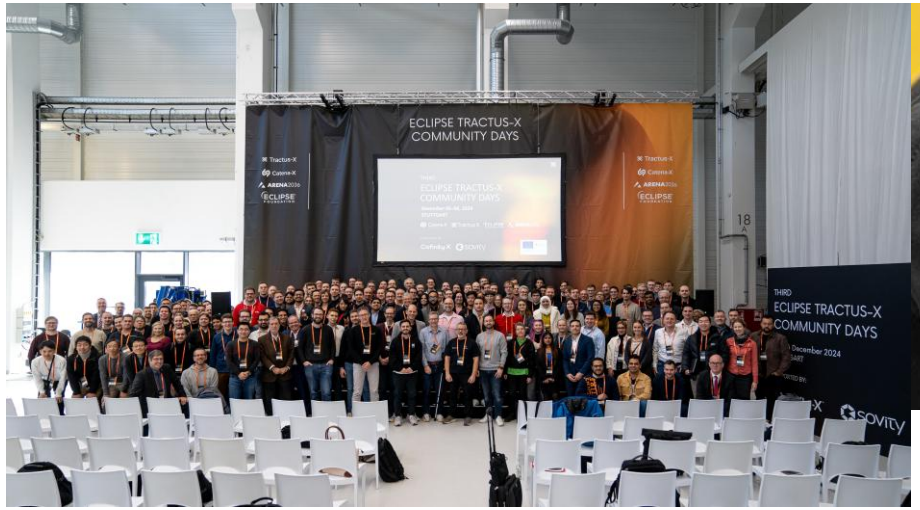
Phase 3 – Join the Community

- Mailing List (tractusx-dev): The formal, legally binding channel for architectural decisions, voting, and release announcements.
- Matrix Space: The primary operational medium for asynchronous technical debugging and peer support.
- Open Office Hours: Weekly "Office Hours" and physical Community Days are essential to establish technical visibility and progress within the project's hierarchy.
- Tractus-X Community Days: Bi-annual physical working conferences where core maintainers and industry adopters rigorously negotiate architectural roadmaps and technical evolution.



https://eclipse-tractusx.github.io/community/open-meetings





We will continue in 2026:

Sixth ECLIPSE TRACTUS-X COMMUNITY DAYS

at the 02./03. of July 2026

STUTTGART

Register now!

Program of the Eclipse Tractus-X Community Days

DAY 1

09:45 AM	Arrival + Welcome
10:15 AM	Keynote
10:40 AM	Introduction of Streams

Stream 1
Presentations Business Perspective

Stream 2
Workshops
Excercises and Learning

05:30 PM	Developer-Panel
06:00 PM	Wrap Up Day 1
06:05 PM	Evening Event at the ARENA2036

DAY 2

09:00 AM	Tech Talk

Stream 2
Workshops
Excercises and Learning

11:45 AM	Results and Learnings
12:15 PM	Summary: Wrap Up and Closing

Register now!

KI-D@ARENA-X
Supporting Startups in the region of Stuttgart

Gefördert vom Verband Region Stuttgart

Main challenges in Research and Development of AI use cases

AI-Startups

- **Data** in sufficient quantity and quality
- **Computing Power** and Availability of AI Infrastructure for Deep-Tech Startups
- **Access to data spaces** for building new AI models in collaboration with research institutions and industry

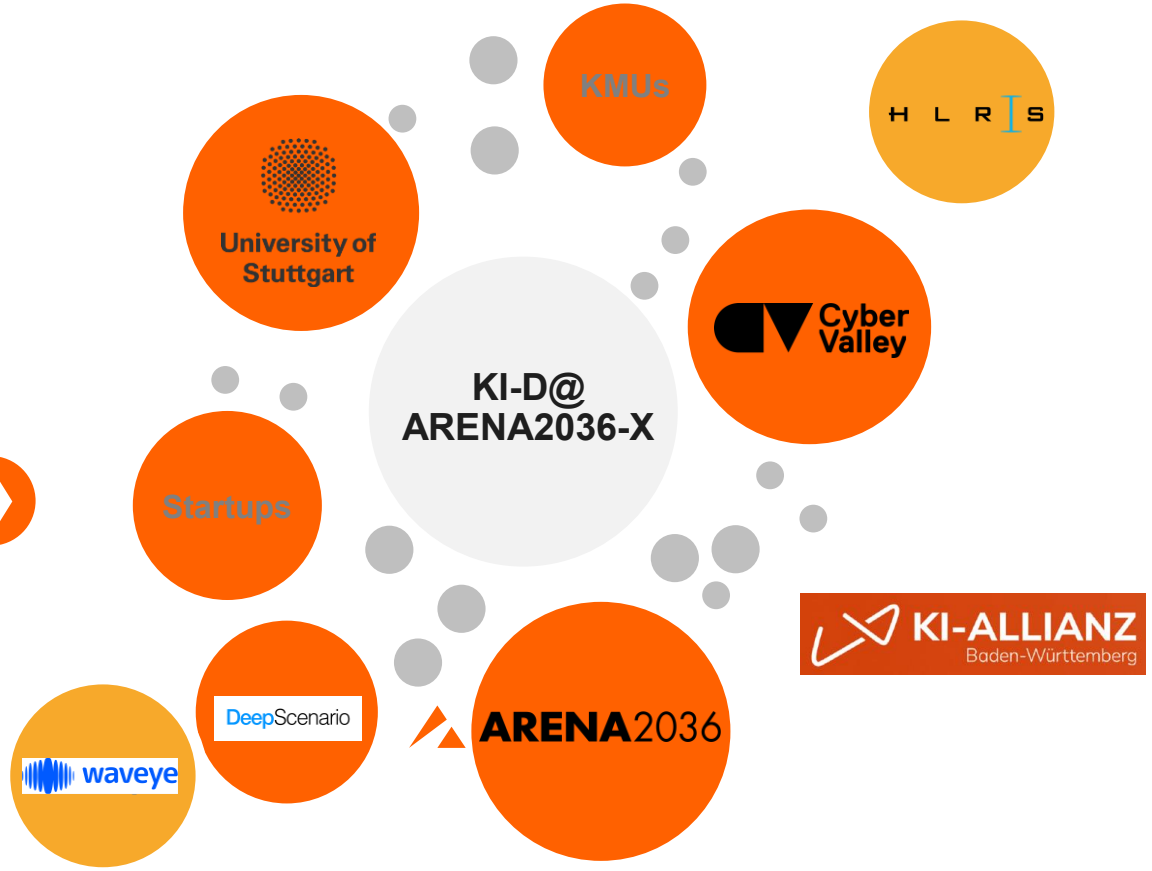
Industry

- There are just a few joint AI projects involving research institutions and innovative startups

Research

- A lack of industrial data spaces, resulting in research that is not **sufficiently application-oriented**

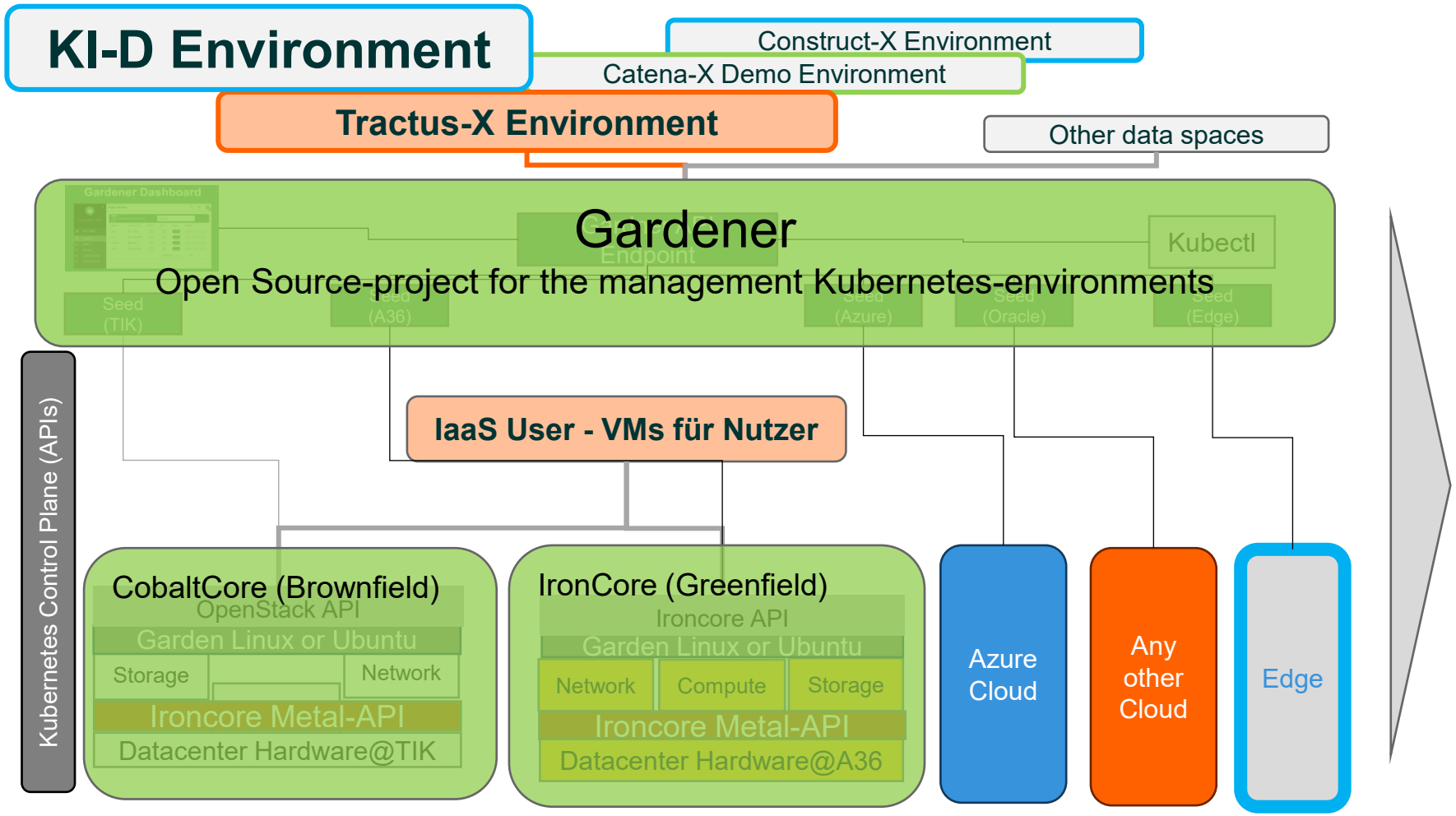
One solution lies in collaboration



A data space-based ecosystem for the development of modern AI systems

ARENA2036

Creation of a federated cloud architecture based on IPCEI-CIS and the project ApeiroRA



Benefits for project partners:

- Access to tractus-X based data spaces
- Access to Kubernetes
- Access to virtual machines
- Easy access to the available infrastructure with computing power für AI use Cases

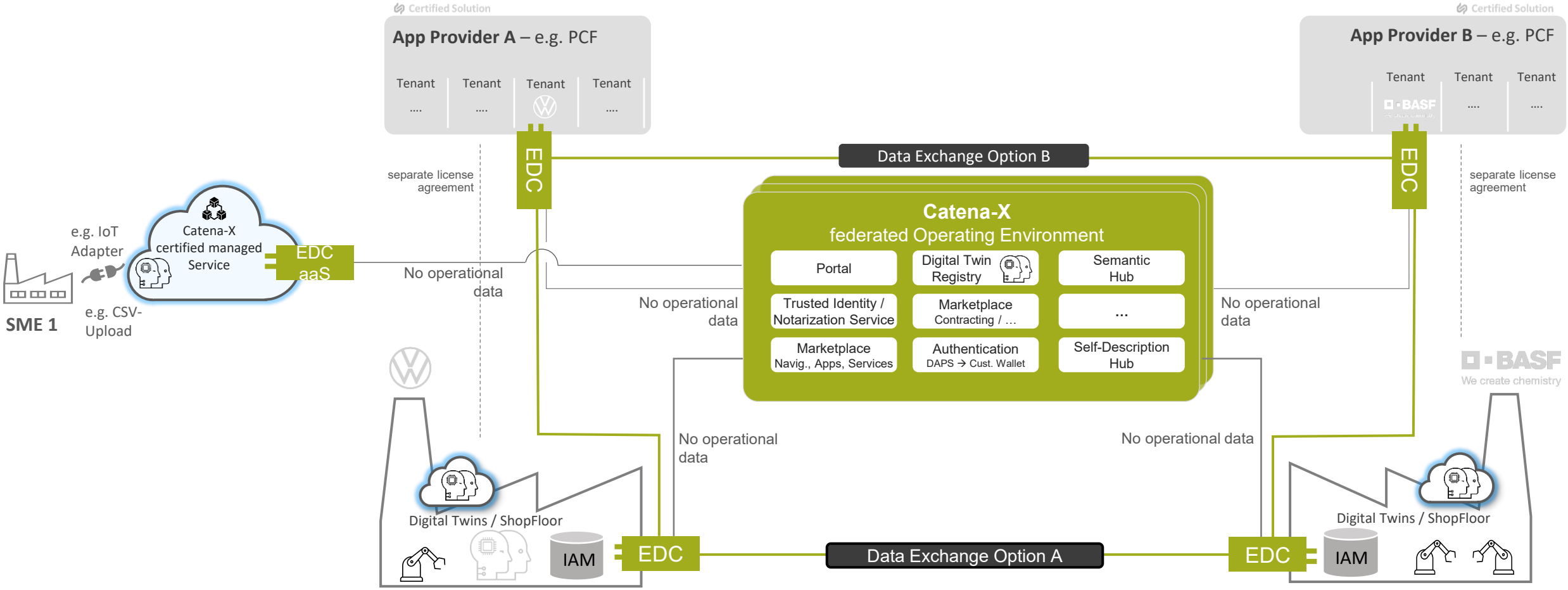
= Open Source Komponente aus dem Projekt AperioRA

Thank you!

Backup

How the Catena-X data space works

- Sovereign, federated and interoperable. Incorporating GAIA-X principals and frameworks.



Note: EDC = Eclipse Data Space Connector (OSS)

ARENA2036

© 2023 Catena-X or a Catena-X affiliate company